

M Sc (BIOTECHNOLOGY)
SEMESTER III EXAMINATION (BACKLOG)
BT802: PLANT AND ANIMAL BIOTECHNOLOGY

Date: 22.04.2011

Time: 10:00 AM to 1:00 PM

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q1. Choose the correct option and put ✓ mark in front of it:

- (1) _____ tissue function's to transport materials through the plant body and provides support
 - a. Dermal
 - b. Vascular
 - c. Ground
 - d. All the above
- (2) The liquid endosperm of coconut contain _____ which will support the continued cell division of mature cells, leading to the formation of callus
 - a. Cytokine
 - b. Auxin
 - c. Giberellin
 - d. All the above
- (3) Asexual reproduction of plants that are considered to be genetically uniform and originated from a single individual or explant is
 - a. Clonal Propagation
 - b. Somaclonal variation
 - c. Callus culture
 - d. Asexual propagation
- (4) Physical transfection method involves
 - a. Microinjection
 - b. Bacterial Transformation
 - c. Lipofection
 - d. None of the above
- (5) Inner cell mass of an embryo is the source of
 - a. Pluripotent cells
 - b. Totipotent cells
 - c. Multipotent cells
 - d. Unipotent cells
- (6) MAS stands for
 - a. Marker associated selection
 - b. Marker aided selection
 - c. Marker assisted selection
 - d. All the above
- (7) Euthanasia means
 - a. Easy killing
 - b. Painless killing
 - c. Mercy killing
 - d. All the above

- (8) Blastulation is formation of
- Solid cell mass
 - Eight cell mass
 - Hollow cell mass
 - All the above
- (9) Exception to the rule of genomic equivalence are
- Lymphocytes
 - Erythrocytes
 - Hepatocytes
 - Melanocytes
- (10) ZP3 glycoprotein play important role in
- Gamete binding
 - Induction
 - Cell differentiation
 - All the above
- (11) Gastrula is a
- Monolayered structure
 - Trilayered structure
 - Bilayered structure
 - Tetralayered structure
- (12) The process of organogenesis starts after the formation of
- Zygote
 - Blastula
 - Morula
 - Gastrula
- (13) Heartwood is the darkened central core of a woody stem.
- ☐ TRUE ☐ FALSE
- (14) Each pollen sac is enclosed by a protective epidermis and a fibrous layer
- ☐ TRUE ☐ FALSE
- (15) Explant is defined as tissue taken from its original site and transferred to an artificial medium for growth or maintenance
- ☐ TRUE ☐ FALSE

Fill in the blanks

- (16) The process of commitment can be divided into specification and _____
- (17) Interaction at close range between two or more cells or tissues of different history and properties is called _____
- (18) Stem cell possess property of self-renewal and _____
- (19) Region of inactive cells behind the root cap is _____
- (20) Phenotypic variation, either genetic or epigenetic in origin, displayed among somaclones is _____

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Part B

Total Marks: 50

Section I

Q1. Write a note on any two:

(10)

- (i) Development of embryo sac
- (ii) Root apical meristem
- (iii) Vascular tissue
- (iv) Causes of dormancy.

Q2. Write a note on any two:

(10)

- (i) Plant tissue culture media
- (ii) Protoplast fusion techniques
- (iii) Callus culture
- (iv) Single cell culture techniques

Section II

Q1. A. Write a note on any one:

(5)

- (i) Gene transfer strategies for plants
- (ii) Ti Plasmid

B. Attempt any one:

(5)

- (i) Explain in brief various gene transfer schemes
- (ii) Write a note on MAS in animals

Q2. Write a note on any two:

(10)

- (i) Process of commitment
- (ii) Induction and competence
- (iii) Genomic imprinting
- (iv) Genomic equivalence

Q3. Write a note on any two:

(10)

- (i) Prevention of polyspermy
- (ii) Gastrulation
- (iii) Spermatogenesis in mammals
- (iv) Structure of sperm